FPQC COVERINGS OF AN ALGEBRAIC SPACE ARE DETECTED AFTER PULLBACK BY A SURJECTIVE ETALE MORPHISM FROM A SCHEME

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Base change of fpqc coverings. lemma-fpqc (carded L26674). The forward direction.
- Scheme-level standard fpqc axioms. Topologies, lemma fpqc-affine-axioms (cross-chapter, named). The assembled family over Z is a standard fpqc covering.
- Openness of surjective etale morphisms of schemes. surjective etale morphisms of schemes are open (cross-chapter, structural). Finite affine opens W_j of U x_X Z cover Z.

Source reference. Original lemma 03MR: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 1523–1523. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03LA: focused-seeds/geometry/S9-topo-recognize-fpqc-covering/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme, and let X be an algebraic space over S. Suppose that $\{f_i : X_i \to X\}_{i\in I}$ is a family of morphisms of algebraic spaces with target X. Let $U \to X$ be a surjective étale morphism from a scheme towards X. Then $\{f_i : X_i \to X\}_{i\in I}$ is an fpqc covering of X if and only if $\{U \times_X X_i \to U\}_{i\in I}$ is an fpqc covering of U.*

Proof. If $\{X_i \to X\}_{i\in I}$ is an fpqc covering, then so is $\{U \times_X X_i \to U\}_{i\in I}$ by Lemma 03MR. Assume that $\{U \times_X X_i \to U\}_{i\in I}$ is an fpqc covering. Let $h : Z \to X$ be a morphism from an affine scheme towards X. Then we see that $U \times_X Z \to Z$ is a surjective étale morphism of schemes, in particular open. Hence we can find finitely many affine opens $W_1, \ldots, W_t$ of $U \times_X Z$ whose images cover Z. For each j we may apply the condition that $\{U \times_X X_i \to U\}_{i\in I}$ is an fpqc covering to the morphism $W_j \to U$, and obtain a standard fpqc covering $\{W_{jl} \to W_j\}$ which refines $\{W_j \times_X X_i \to W_j\}_{i\in I}$. Hence $\{W_{jl} \to Z\}$ is a standard fpqc covering of Z (see Topologies, Lemma 03LA) which refines $\{Z \times_X X_i \to X\}$ and we win. □

Date: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.